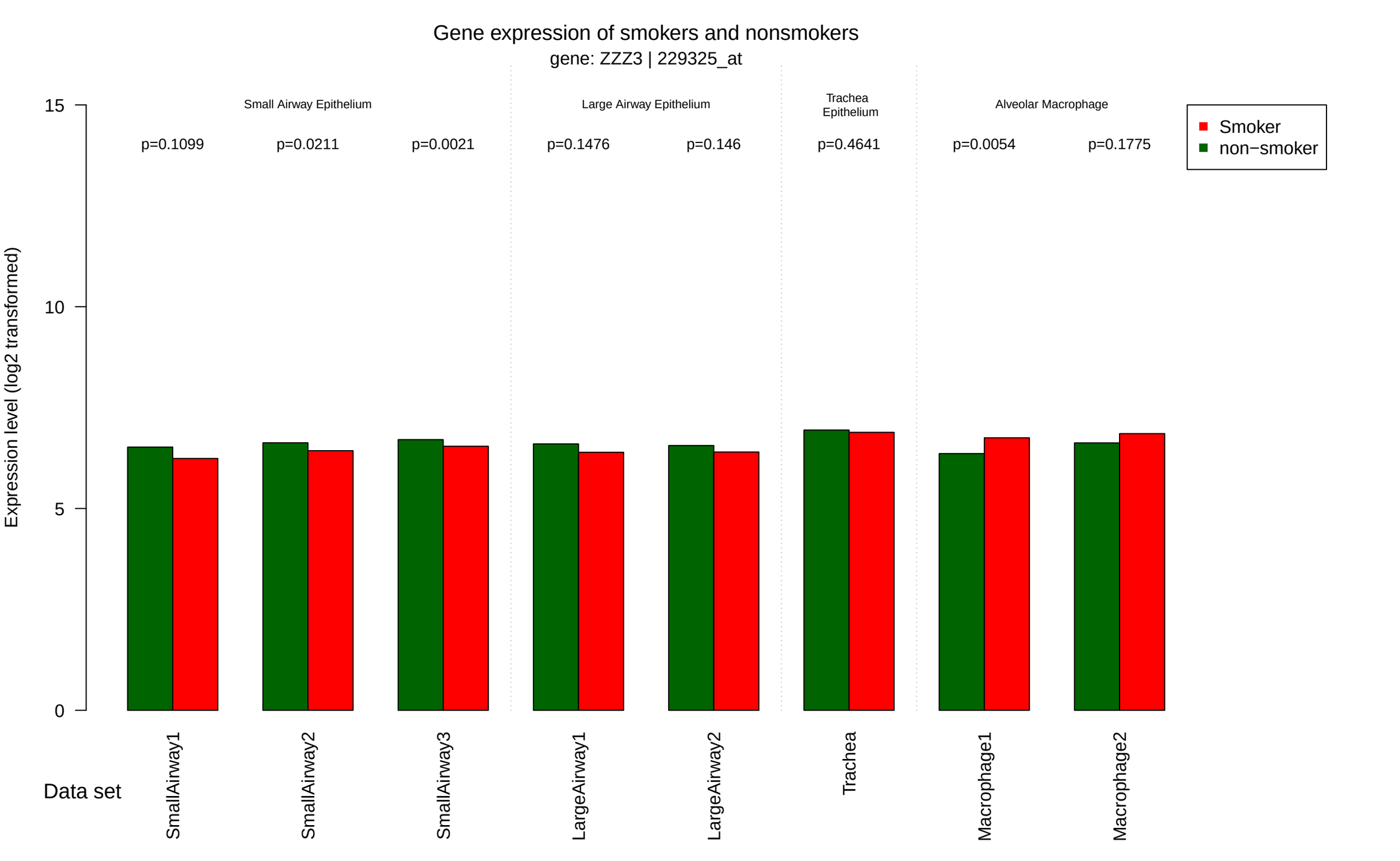
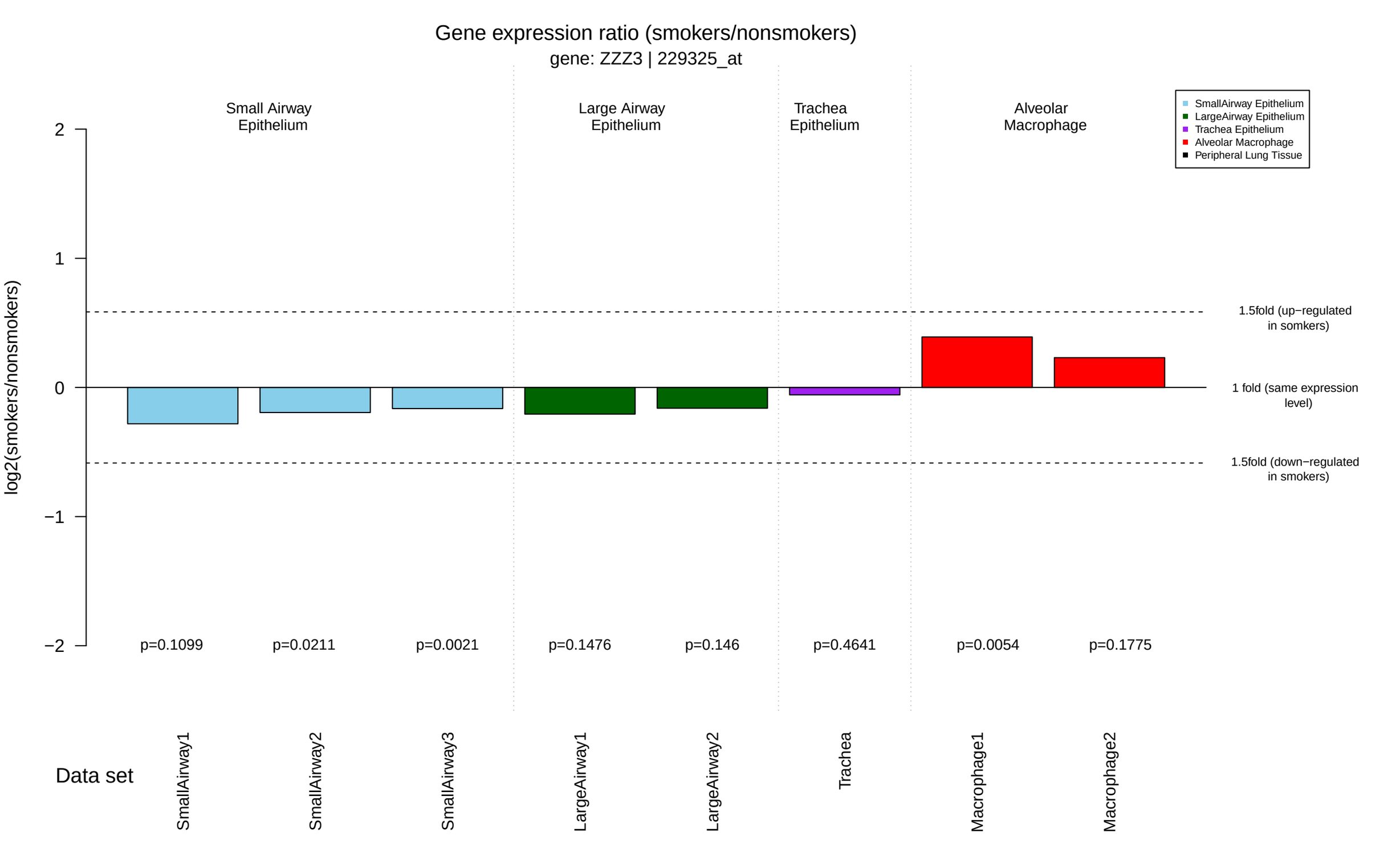
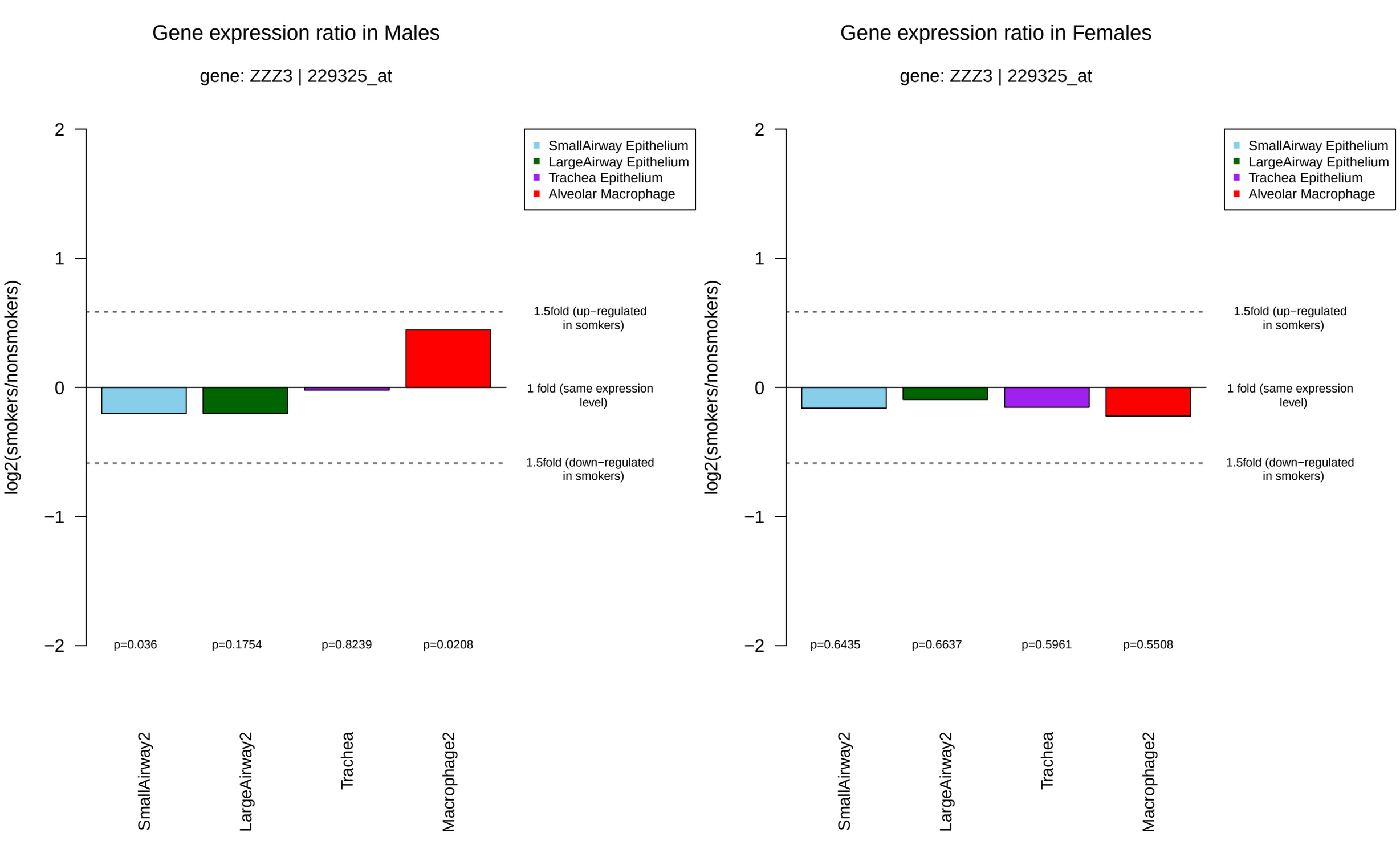
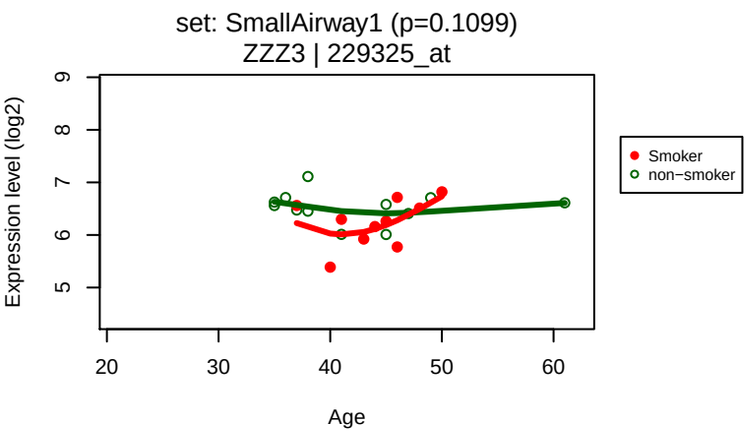
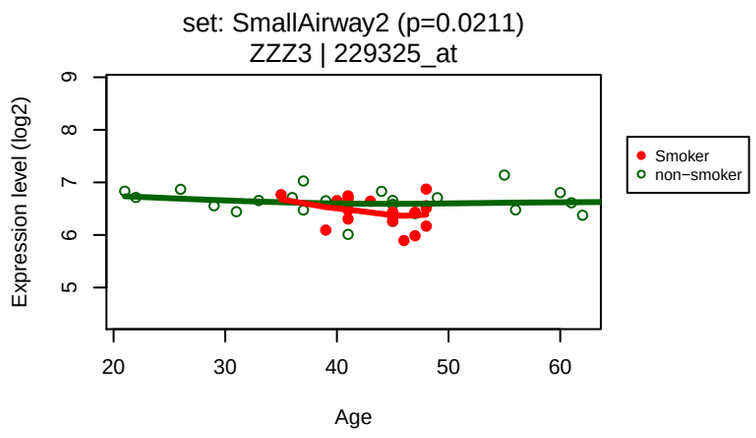
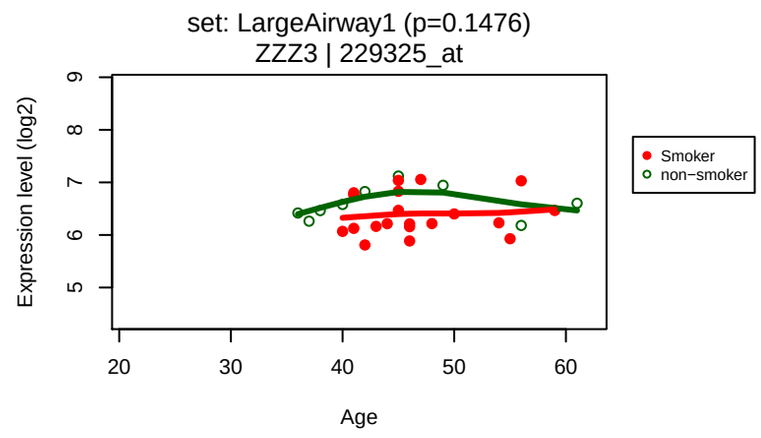

Gene expression of smokers and nonsmokers
gene: ZZZ3 | 229325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1099
p=0.0211
p=0.0021
p=0.1476
p=0.146
p=0.4641
p=0.0054
p=0.1775
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

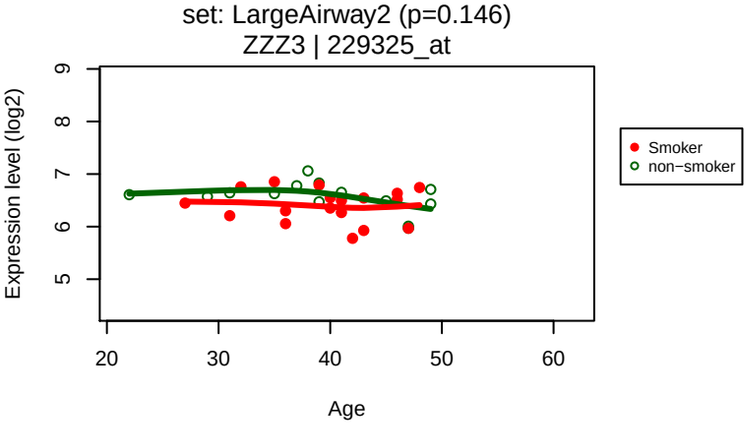

Gene expression ratio (smokers/nonsmokers)
gene: ZZZ3 | 229325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1099
p=0.0211
p=0.0021
p=0.1476
p=0.146
p=0.4641
p=0.0054
p=0.1775
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

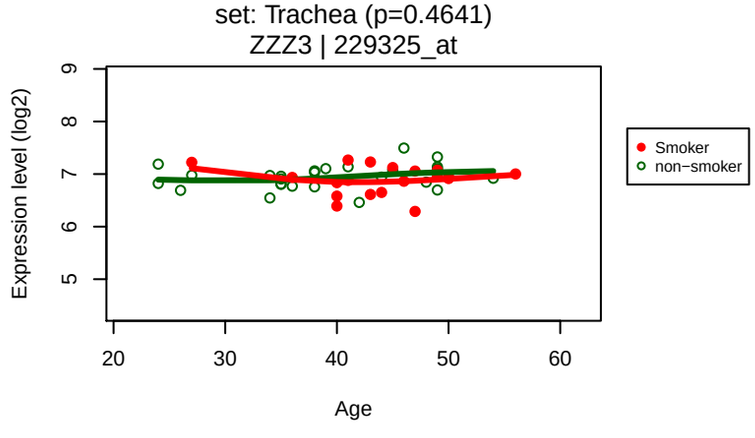

Gene expression ratio in Males
gene: ZZZ3 | 229325_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.036
p=0.1754
p=0.8239
p=0.0208
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZZZ3 | 229325_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6435
p=0.6637
p=0.5961
p=0.5508
SmallAirway2
LargeAirway2
Trachea
Macrophage2

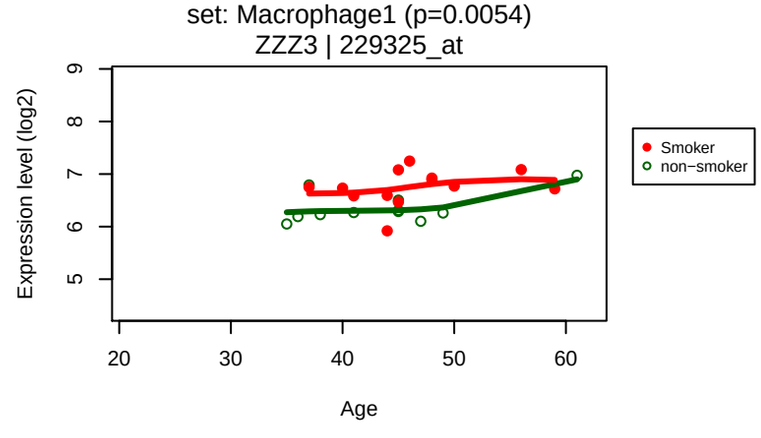

set: SmallAirway1 (p=0.1099)
ZZZ3 | 229325_at
Expression level (log2)
Age
Smoker
non−smoker

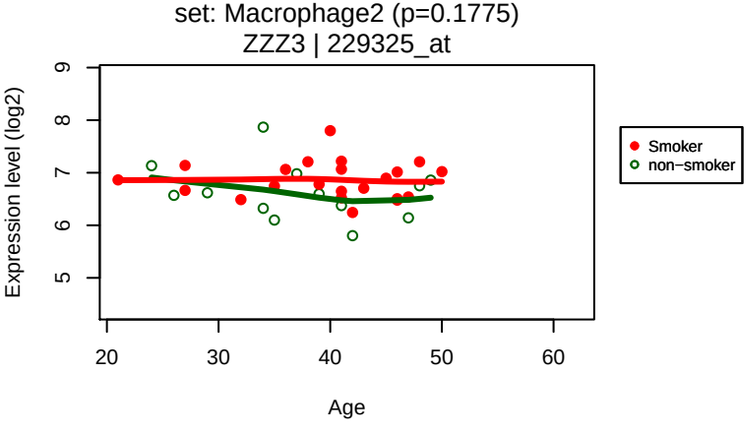

set: SmallAirway2 (p=0.0211)
ZZZ3 | 229325_at
Expression level (log2)
Age
Smoker
non−smoker

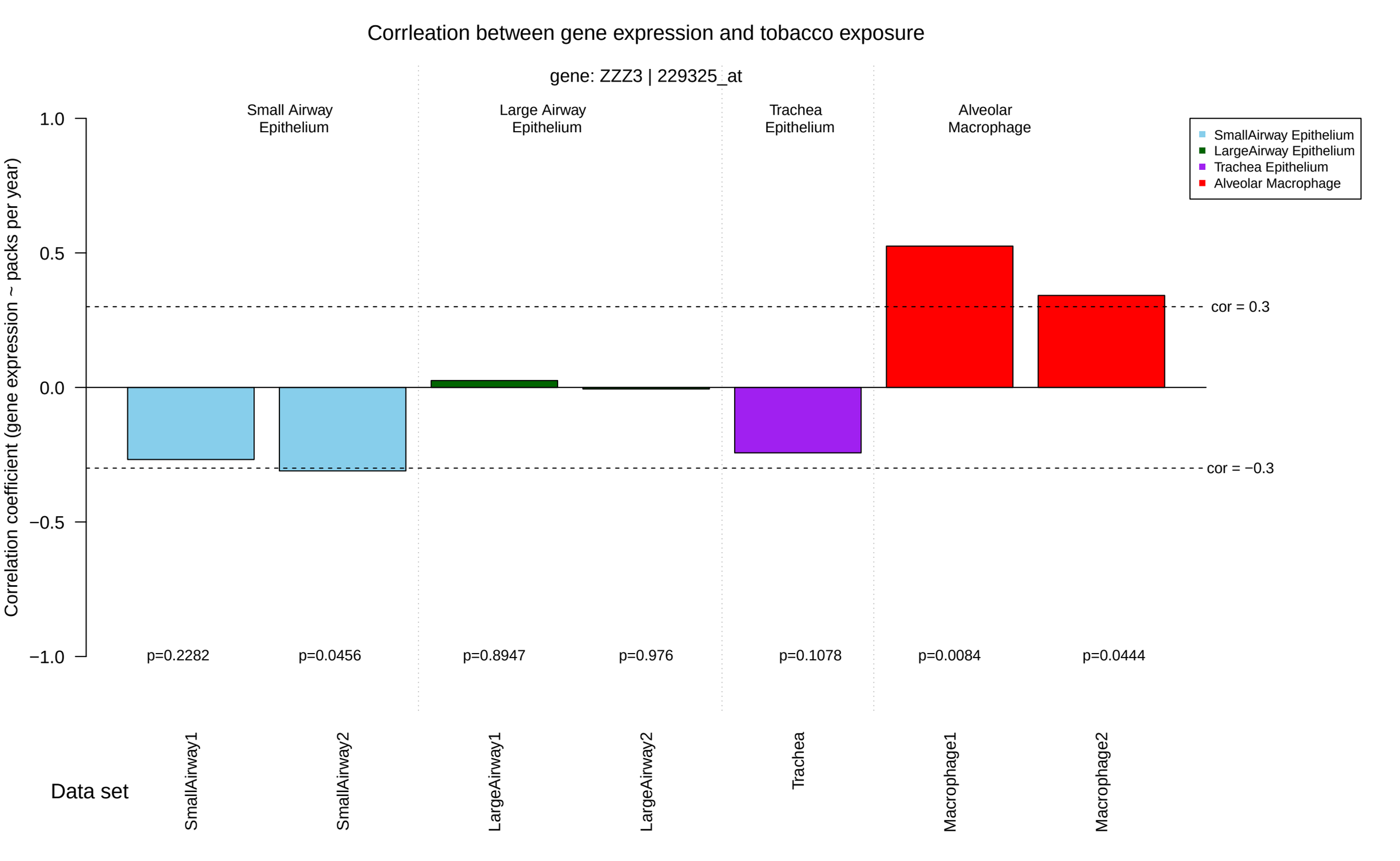

set: LargeAirway1 (p=0.1476)
ZZZ3 | 229325_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.146)
ZZZ3 | 229325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4641)
ZZZ3 | 229325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0054)
ZZZ3 | 229325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1775)
ZZZ3 | 229325_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZZZ3 | 229325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2282
p=0.0456
p=0.8947
p=0.976
p=0.1078
p=0.0084
p=0.0444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2